Abstract

Partial Differential Equations (PDE's) can be notoriously difficult to solve. A popular approach to solving this problem is by using physics-informed neural networks (PINN's) which leverage the ability of neural networks as universal function approximators by adding the PDE constraints in the loss function. Our work investigates whether a Physics-Informed Quantum Neural Operator (Q-PINO) can be designed so that its data-embedding circuit is matched to the spectral structure generated by a nonlinear PDE. Variational quantum models can be expressed as Fourier-type sums whose accessible frequencies are determined by the eigenvalue gaps of their data-encoding Hamiltonians. We choose the Fisher–KPP equation for its nonlinearity, reduce it into a finite set of interacting spectral modes, and analyze how diffusion, growth, and nonlinear mode coupling shape the resulting dynamics. The goal is to use this PDE spectral action to reverse engineer encoding Hamiltonians whose accessible quantum feature frequencies (the eigenvalues of the Hamiltonian) are naturally aligned with the dynamics the Q-PINO must learn. This provides a physics-guided approach to quantum circuit design in which the embedding architecture is chosen from the structure of the governing PDE rather than selected arbitrarily.